Claims data that reconcile.

A reproducible Python and SQL workflow over CMS synthetic inpatient claims. Annual membership: CLM_THRU_DT in 2008. No real patient or Kaiser data.

116,352

Synthetic beneficiaries in the full ledger

27,449

Consistent claims in complete domain

Complete-domain claims: 27,449 across 116,331 nonambiguous beneficiaries. Another 21 beneficiaries remain in the ledger but are excluded from complete summaries.



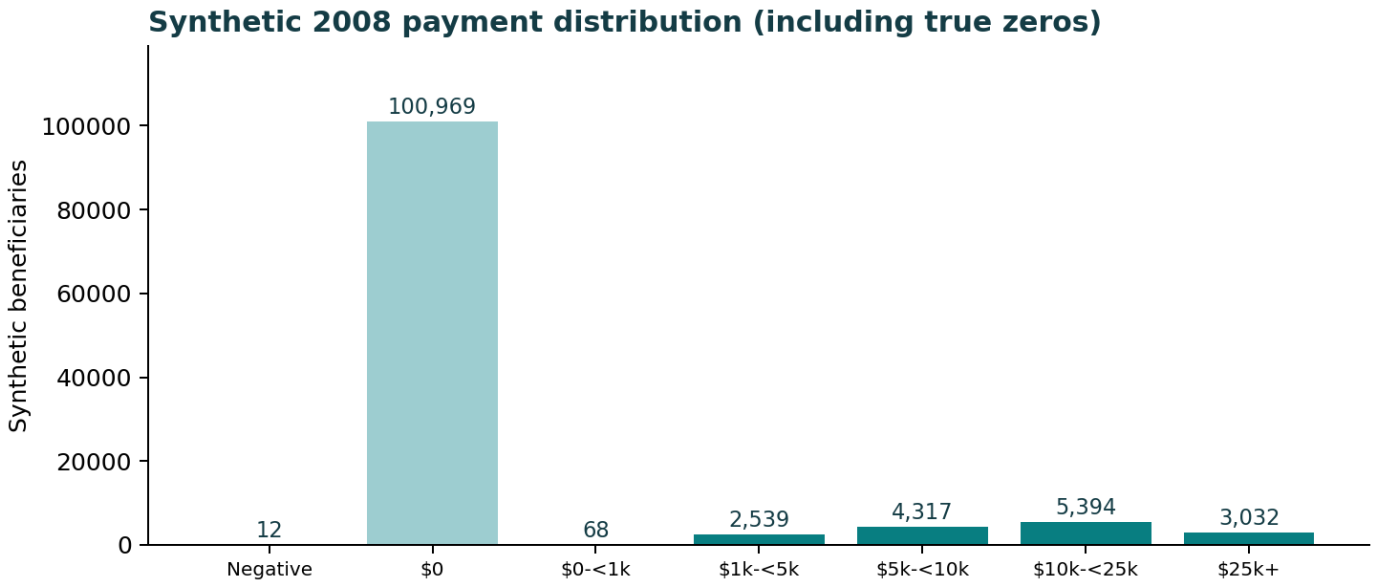
SYNTHETIC · Sample 1 · Claim amounts counted once only after conflict checks

Grain check. 66,773 source rows represent 66,705 claim keys. Of 68 multi-segment claims, 67 have conflicting fields and are quarantined. Claim-level billing dates can be recovered from consistent nonmissing segment values; conflicting payments are never summed or chosen arbitrarily.

Annual flow. 27,496 original claim keys end in 2008; 21 are quarantined. The remaining 27,475 accepted claims link many-to-one to the beneficiary table without row or payment inflation. Excluding all accepted claims belonging to the 21 affected people yields the complete-domain 27,449 claims.

Descriptive result. 15,724 of 116,331 nonambiguous synthetic beneficiaries (13.52%) have an accepted claim. There are 100,607 true zero-claim beneficiaries. These are sample descriptions, not population or annual incidence estimates.

Payments, coverage and limits.



Denominator. Among 116,331 nonambiguous people, including true zeros, mean synthetic claim payment is $2,185.66; median $0; p90 $6,000. The 12-month Part A domain has 104,281 people and claimant share 14.68%.

Exact reconciliation. Independent DuckDB reconstructs claim grain from staged segments; SQLite separately checks canonical annual claims. Sixteen count/cent identities pass. The official expanded formula, payment + covered days x pass-through per diem, exactly reproduces MEDREIMB_IP for all 116,331 paired nonambiguous beneficiaries. Payment-only sums fall short by $2,733,780.

Limits. Fully synthetic data for software development and training, not real patient records or reliable Medicare population inference. Claim counts are not hospitalization episodes. The annual rule is claim-through date, not admission or payment year. Partial coverage and death can limit observation; counts per beneficiary are not full-year incidence rates. Conflicting claims are quarantined and affected beneficiaries excluded from complete-analysis summaries; exclusions can change the descriptive distributions. Synthetic dates, covered days and amounts must not be altered to make medical patterns look realistic. No hospital ranking, causal conclusion, cost saving or clinical recommendation is supported.

Sources. CMS DE-SynPUF Sample 1 beneficiary/inpatient ZIPs; codebook IP-3/IP-7, Appendix 2, and user manual. Exact URLs and SHA256: docs/source_lock.json. Published outputs contain aggregates only.